

Artificial Intelligence Ethics

-

A Question of Moral Law and Ethical Standards?



[National Science Foundation](#) on [02/06/2025](#)

Foreward

While there has been considerable research, approaches, and some implementations of “Values” based programming within Artificial General Intelligence (AGI), the current methodology that I have reviewed in my research does not establish a clear standard for Moral Law or Ethics-based Logical parameters.

There appear to be many disparate forms of implementing code and systems to review any moral or ethical issues surrounding the effectiveness of Artificial Intelligence applications within AGI. (With AGI being defined by [Amazon Web Services \(AWS\)](#)¹ as [Artificial general intelligence \(AGI\)](#) being a field of theoretical AI research

that attempts to create software with human-like intelligence, the ability to self-teach, and that AGI is a theoretical representation of a complete artificial intelligence (AI) that solves complex tasks with generalized human cognitive abilities.

Also, defined by AWS, Artificial Intelligence (AI) is AI technologies that function within a set of pre-determined parameters.¹)

This document proposes the consideration of establishing a set of Ethics-Based Logical parameters for AGI and AI.

As stated in a research paper³, Titled, [Investigating machine moral judgement through the Delphi experiment, January 13, 2025](#), the “Computational Morality” being proposed is based upon philosophical ideals of John Rawls, Kant (deontology), and the Common Sense Norm Bank (Norm Bank). In addition, as stated in the Delphi experiment paper, the computational framework for “values” are subject to predictive moral judgements and in accordance with the work of Rawls, to be based upon “patterns of human ethics via crowdsourcing moral opinions”.

There are other concerns related to the use of Retrieval Augmented Generation (RAG) related to AI Ethics which only solidifies my opinion, which is that further consideration should be given to establish a “Moral Law” or an Ethical Framework Standard on which any AGI / AI code should be built. (See this article, [“Ethics and Accountability in Human-AI Collaboration Using RAG AI”](#) by Abhinav Aggarwal, dated 11/26/2024 for additional information on RAG and the Ethical concerns)

My research and subsequent conclusions are not exhaustive, but it does include articles/information from the following:

- The work of Nick Bostrom; specifically on “SuperIntelligence” (<https://nickbostrom.com/>)
- Various Articles, Talks by Yejin Choi, Professor and Senior Fellow, Stanford University.²
- Stanford University Human Centered Artificial Intelligence (Stanford HAI - <https://hai.stanford.edu/>)
- Allen Institute for AI (<https://allenai.org/>)
- ChatGPT (<https://chatgpt.com/>)
- U.S. National Science Foundation, [Ethical and Responsible Research](#)
- See [References/Bibliography page](#) for other resources.



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It is not the machines (AI/AGI) we should fear, but those that create and control the machines that have no ethical standard or moral law.

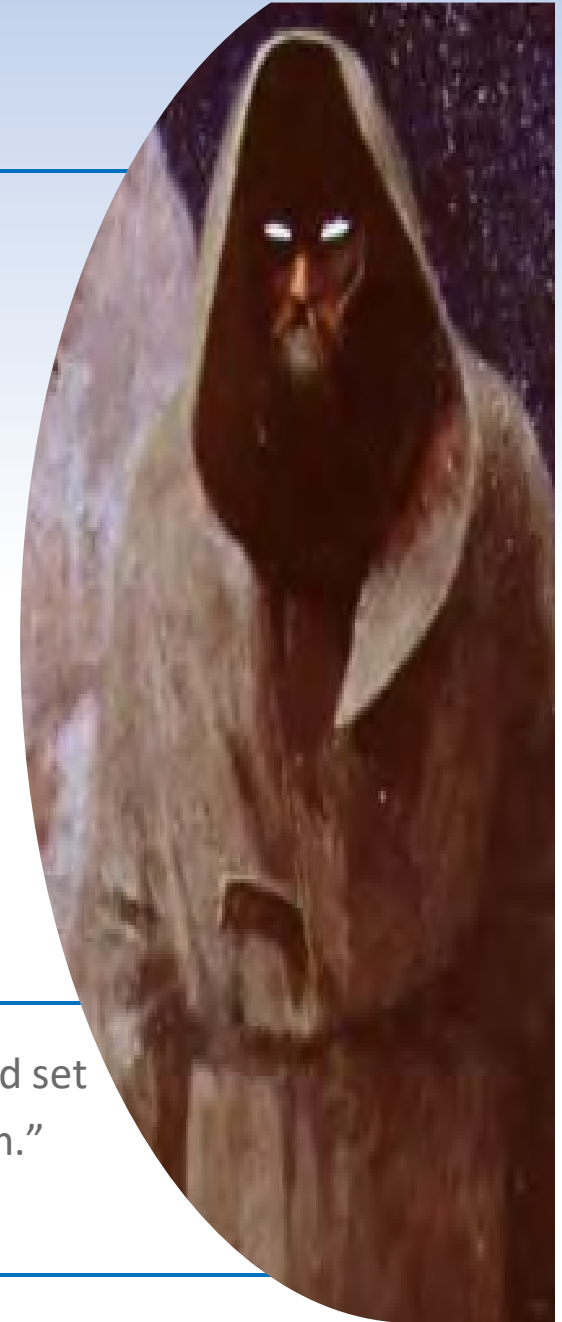
For further clarification of this point, Ignatius Cox, S.J., Ph.D. of Fordham University, states in, "Liberty Its Use and Abuse" ¹⁰:

- It is universally recognized by men that there is a difference between right and wrong in man's free or volitional actions, that in consequence there are some actions which man ought not to perform, some which he ought to perform, some which he may perform or omit just as he pleases. This universal judgment (recognition) of men is expressed very frequently in propositions such as these: "The moral law forbids such an action." "The moral law commands such an action." "The moral law permits such an action"

If a moral law, a "values" system, can be defined outside of man's determinate subjective "values" system, then programming AGI to understand a standard moral law is also possible. If instead, a subjective "values" system is utilized to formulate and build AGI ethics, the premise on which this "values" system is built will always be inherently flawed as it will be subjective to the morality of human nature.

"Once men turned their thinking over to machines in the hope that this would set them free. But that only permitted other men with machines to enslave them."

-Frank Herbert, Dune



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AI ETHICS ARTICLE. In the article titled, [*Reflections on Putting AI Ethics into Practice: How Three AI Ethics Approaches Conceptualize Theory and Practice*](#), by Hannah Bleher · Matthias Braun, they examine some of the current (2023) approaches to AI ethics. Specifically three approaches to applied AI ethics: the embedded ethics approach, the ethically aligned approach, and the Value Sensitive Design (VSD) approach.

This image (Fig. 1) reflects the article's defined "Meta-Framework. This framework is the three Conceptual Dimensions for Applied AI Ethics (from the Bleher and Braun 2023 article), in which the "first dimension of reflection focuses on affects/emotions, the second dimension on justifications, and the third dimension asks about governance aspects. These dimensions are connected to the guiding elements that drive the three analyzed approaches: intuitions, principles, and deliberation." ⁴

OBSERVATIONS. The research paper reviews the current approaches to AI Ethics in practice. There are mentions by the authors of concern related to bias in the definition of "principles"; i.e., principles are subject to the person(s) defining the principle. Their defined "Meta-Framework" for Applied AI Ethics approaches (See Fig. 1) have shortcomings outlined in their paper ([page 15](#)) with concerns of "affect" and emotion, moral attitudes, biases, and subjectively defined principles. In summary, there seems to be disparate opinions from this and other research on the prevailing approach to AI Ethics.



Fig. 1 - Meta-Framework for Applied AI Ethics Approaches and its Three Conceptual Dimensions – by Hannah Bleher · Matthias Braun ⁴

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Summary of a “Norm” based approach to AI Machine Ethics:

Related to the development of Moral Values or Ethical Standards, the Allen Institute has taken a stance that does not define specific “Moral Laws” or standards. For clarification, [The Allen Institute](#) is non-profit AI research institute founded in 2014 by the late Paul Allen (Microsoft Co-founder). Their [website](#) states the following: “we develop foundational AI research and innovation to deliver real-world impact through large-scale open models, data, robotics, conservation, and beyond.”

A researcher, from The Allen Institute states in a Nov. 3, 2021, IEEE Article by Charles Q. Choi entitled, “[Machines Learn Good From Commonsense Norm Bank New moral reference guide for AI draws from advice columns and ethics message boards](#)”⁵, that, “AI systems are being entrusted with increasing authority in a wide range of domains—for example, screening resumes [and] authorizing loans,” says study co-author Chandra Bhagavatula, an artificial intelligence researcher at the Allen Institute for Artificial Intelligence. “Therefore, it is imperative that we investigate machine ethics—endowing machines with the ability to make moral decisions in real-world settings.”

The question of how to program morals into AIs goes back at least to Isaac Asimov's Three Laws of Robotics, first introduced in his 1942 short story "Runaround," which go as follows:

1. A robot may not injure a human being or, through inaction, allow a human being to come to harm.
2. A robot must obey the orders given it by human beings except where such orders would conflict with the First Law.
3. A robot must protect its own existence as long as such protection does not conflict with the First or Second Laws.

In the IEEE article, there is a statement by another Allen Institute researcher and study co-author, Ronan Le Bras, in which states, “we decided to approach this work from the perspective of descriptive ethics—that is, judgments of social acceptability and ethics that people would make in the face of everyday situations,”⁵.

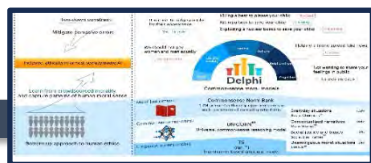
OBSERVATION: The Allen Institute’s approach to Machine Ethics seems to be one that is a gathering of opinions (“Norms”) on which to form AGI/AI moral values. While this seems to appease some with the statements of “descriptive ethics”; it is not considering the issue that without a standard or “truth”, this type of ethics become subjective over time. My observation is that logic would dictate that the AGI/AI (LLM) programming code could be complex variations of the the “old-school” simplistic BASIC programming, “If/then And/But” statements with infinite possibilities and variables. As with the simplistic IF/THEN code there could be in the formation of any AGI/AI code programming a set of STANDARD variables, “Moral Law”, or Truth statements. From these set of Ethical Standards or “Values Based Framework” the rest of the code could be built.



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Examples of additional approaches to AI/AGI Ethics - Summaries

In 2021, researchers at the University of Washington and the [Allen Institute for Artificial Intelligence \(AI2\)](#) in Seattle developed and introduced Delphi, an experimental framework based on deep neural networks trained directly to reason about descriptive ethical judgments. In an article by Professor Derek Leben of Carnegie Mellon University in Jan. 2025, he provides additional detail regarding this approach to AI Ethics by stating, *"They used a large language model (LLM) to train Delphi on a database of 1.7 million ethical judgments, hoping to crowdsource moral reasoning and bring more clarity to AI ethics."*⁷ Professor Leben also states, this insight of the postmortem analysis of Delphi by philosophers Colin Allen and Brett Karlan who, he states, correctly observe:...the problems [with Delphi] go deeper than a lack of hindsight and foresight. Delphi demonstrates a kind of disciplinary hubris and lack of basic scholarship that afflicts too much (but not all) of the work by computer scientists in the expanding arena of 'AI ethics.'⁷ The article provides an examination of Delphi's issues in making ethical judgements, and discusses other approaches to computational morality; however, there still seems to be a lack of clarity on how to proceed in the area of AGI/AI Ethics.



In the theses, ["Avoiding the Basilisk: An Evaluation of Top-Down, Bottom-Up, and Hybrid Ethical Approaches to Artificial Intelligence"](#) by Cole Shardelow (2021)⁶, he thoroughly examines current approaches to the implementation of morals within AI (Super AI or AGI).

The theses begins with a reference to "a popular 2010 internet thought experiment known as Roko's Basilisk which was proposed on the philosophy and artificial intelligence forum *Less Wrong*". It then continues to present arguments for and against the Bottom-Up, Top-Down, and Hybrid Ethical approaches to the issue of Ethics within AI. It is a well-researched document, and has detailed analysis of these approaches. In the conclusion, the author suggests the Hybrid approach would be optimal for the implementation of ethics within AI to allow for the integration of moral principles. This would allow for an AGI/AI to learn from moral consensus as situations and applications for the use of AI evolve. In addition, I would agree with the author's conclusion in that there does not seem to be an agreed upon approach to AI Ethics.



Basilisk image created using Canva.com

In an excellent paper by Benjamin Kuipers from 2016, titled, ["Human-like Morality and Ethics for Robots"](#), he elaborates on Robots (ultimately AI) operating at some point without being controlled by humans. His paper continues to describe an approach to the integration of moral and ethical principals within Robots and AI. He incorporates a variety of examples of scenarios that are worth examining in terms of the dilemma that humanity faces when considering the ethical implications of AI. The information in his paper includes the consideration of the defined use of "Utilitarianism ("the greatest good for the greatest number") and deontology ("duties and rules specifying what is right and what is wrong") which are often considered opposing alternative positions on the nature of ethics.

These ethical philosophical concepts are utilized in other current approaches to the issue of Ethics within AI, and have some merit for being foundational in AI Ethics.

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CONCERNS:

Consideration of RISK and Misuse in regards to the continuation of AGI/AI development without an industry agreed upon Ethical Standard.

WISDOM

dictates, that in an effort to win at innovation and monetization, we need not **fail** in the area of an AGI/AI ETHICAL industry standard; as it will be to our detriment if we do not seriously consider the destructive implications to neglect the issue for the sake of technological progress.

CONSIDERATION

of the possibility to continue reviewing the conceptual framework of the OSI model or other standards based technology concepts. Specifically, related to the OSI Model, the Application Layer (7) standard might be considered to incorporate an Ethical-based standard for AI/AGI?

Mapping an Artificial General Intelligence (AGI) to a Large Language Model (LLM) using the OSI model would conceptually place the AGI at the "Application Layer" (Layer 7), interacting with the LLM as a specialized application that leverages the underlying network capabilities provided by the lower layers, including data transmission, routing, and session management, which would be handled by the LLM

In an article titled, "[Driving Adoption: Using the OSI Model to Evaluate Large Language Models' Readiness](#)" on LinkedIn from May 16, 2023, Bimal Maharjan of Gutsphere, Inc.⁹, the author presents an OSI (Open Systems Interconnection) metaphor for LLM Market Adoption Readiness.

The premise being to utilize the OSI model, or Open Systems Interconnection model, (which was officially established in 1984 by the International Organization for Standardization (ISO) as the 7-layer conceptual framework/standard for network communications), to review AI's LLM (Large Language Model) Market Readiness.

OSI Metaphor for LLM Market Adoption Readiness⁹

OSI Layer	LLM Equivalent	Status
Physical Layer	Server infrastructure to run LLMs	Ready
Data Link Layer	Software connecting the physical infrastructure to the LLM	Ready
Network Layer	Ability to distribute and access the LLM service globally via the internet	Ready
Transport Layer	Reliability of the LLM service, including uptime, error handling, etc.	Ready, with room for improvement
Session Layer	User interaction with the LLM, including session management and context	Needs improvement
Presentation Layer	User Interface (UI) and User Experience (UX) with the LLM	Needs improvement
Application Layer	Specific applications of the LLM (chatbots, virtual assistants, etc.)	Needs significant development



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In the GOOGLE DEEPMIND Article: “Mapping the misuse of generative AI” from August 2, 2024 by Authors Nahema Marchal and Rachel Xu, they state that: By analyzing media reports, they identified two main categories of generative AI misuse tactics: the exploitation of generative AI capabilities and the compromise of generative AI systems.⁸ See Fig.1, Fig 2. <https://deepmind.google/discover/blog/mapping-the-misuse-of-generative-ai/>

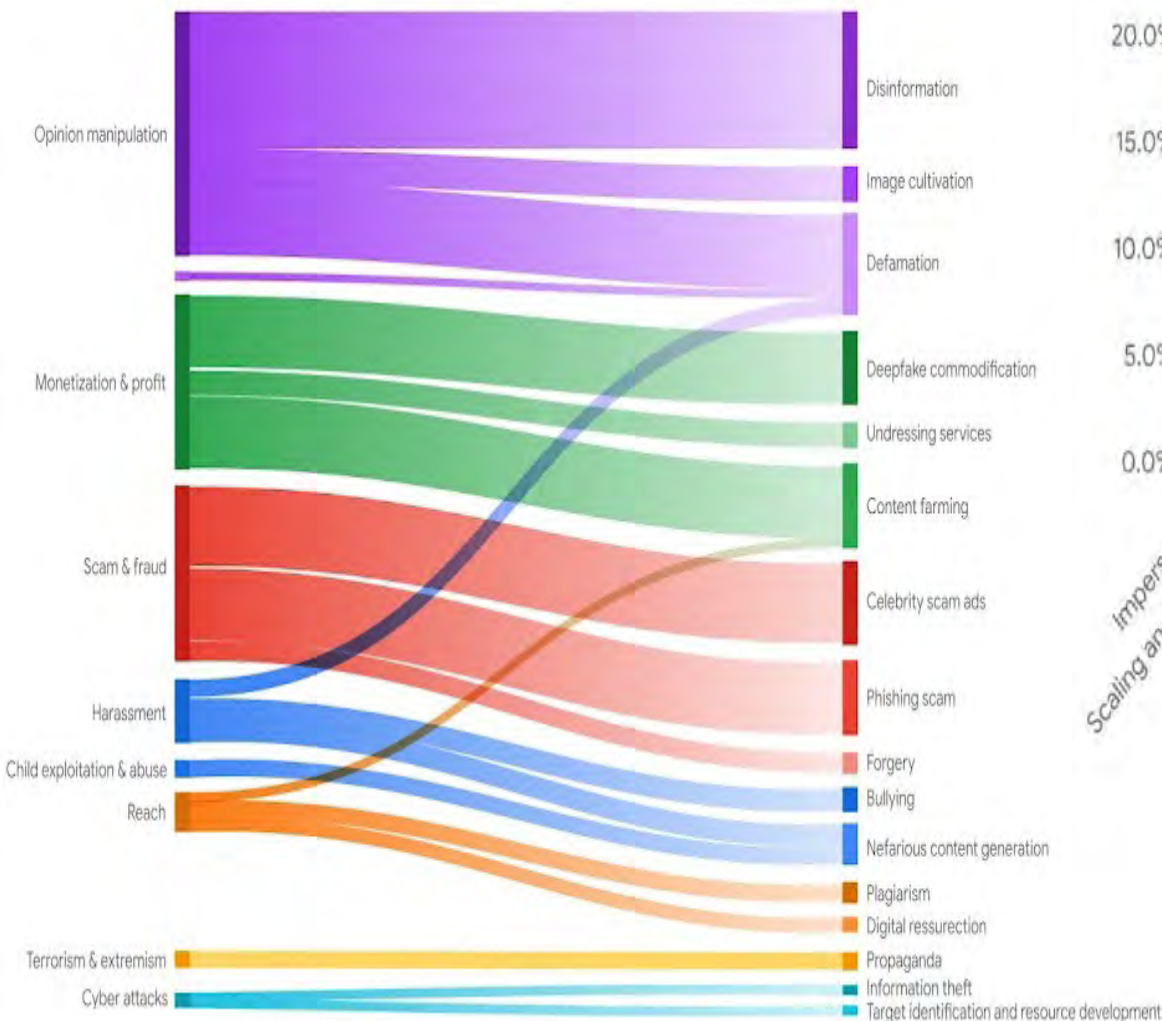


Figure 2

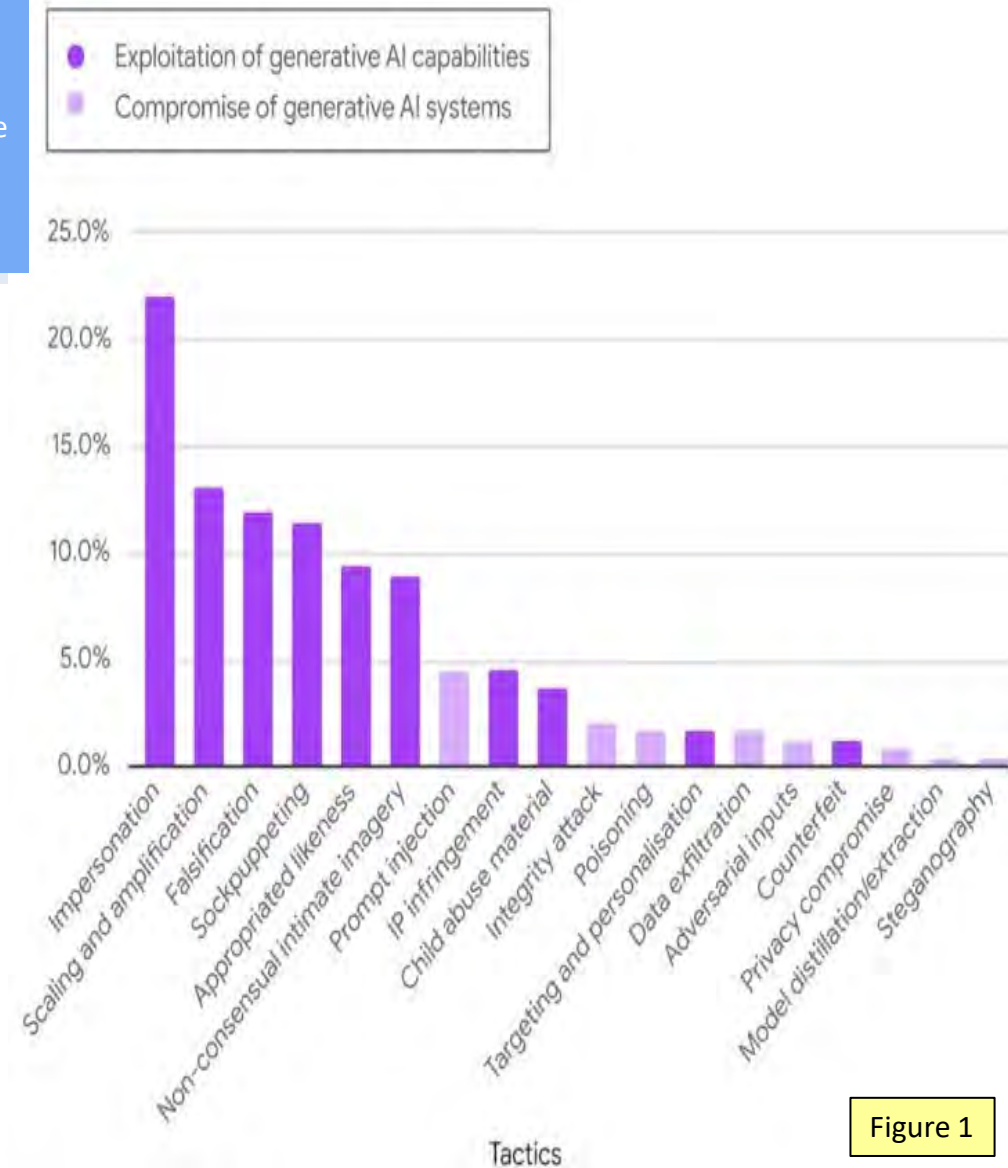


Figure 1

Figure 2, (from the Google Deepmind Article, “Mapping the misuse of generative AI”), the diagram shows of how the goals of bad actors (left) map onto their strategies of misuse (right).⁸

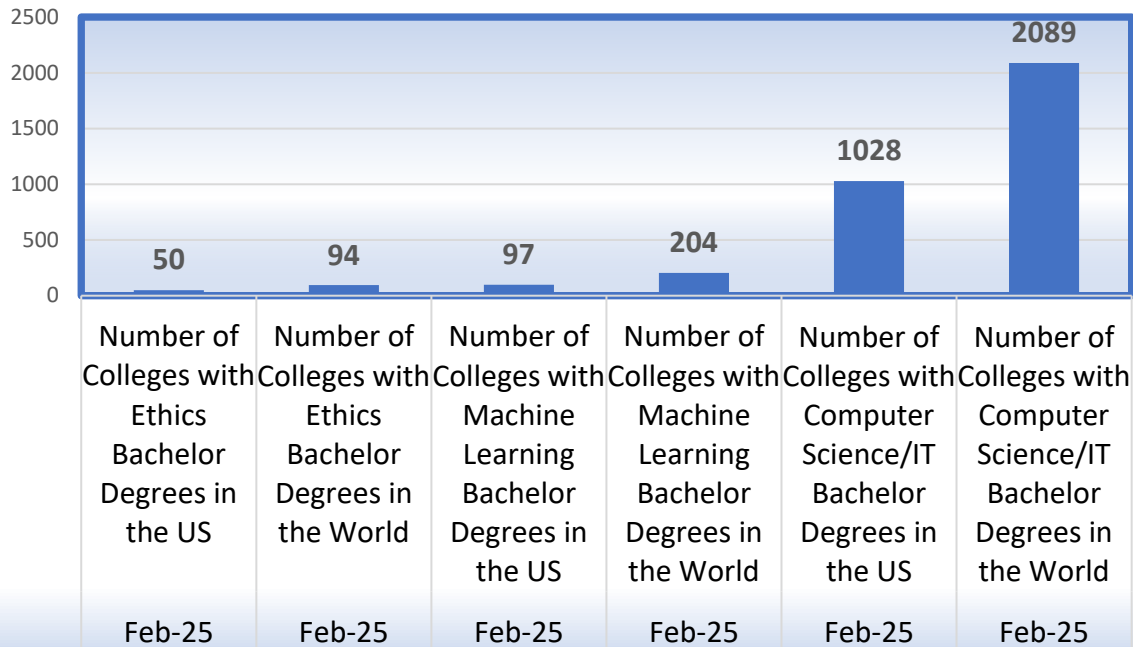
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CONSIDERATION: Expansion of ETHICS Course requirements for Computer Science, AI, and Machine Learning at College or High School Levels.

[Bachelorsportal.com](https://bachelorsportal.com) reveals the number of colleges both in the World and US that currently offer an Ethics Bachelor’s Degree versus the number of colleges that offer Computer Science or Machine Learning Bachelor’s Degrees. While some school programs do include an Ethics class as part of a Core Study, it seems that in the definition of “Ethics” or “Truth” there is more concern with inclusion, equity and/or bias then actual “Ethics”. (See University of Texas, Computer Science Dept. article, Titled, “[The Importance of Ethics in Computer Science, Posted by Bettina Mateo and Lauren Cotton | February 18, 2022](#)”), which begins with, “As society’s reliance on technology increases, computer scientists are now placed at the forefront of new and unique ethical decisions. However, most computer science curriculums don’t require a substantial education in ethics” ¹¹. There be at minimum the inclusion of Ethics teaching within the Computer Science, LLM, and AI disciplines.

“[Why Universities Should Adopt Artificial Intelligence \(AI\) Ethics Course Requirements](#)”, she discussed with Will Griffin, at the time, the Chief Ethics Officer at Hypergiant, as to the need for Ethics to be taught in Computer Science and Engineering schools within Universities. The article states that there are some schools that currently teach ethics; however, they are not mandatory: (this list is as of 2022) - Stanford, Harvard, Carnegie Mellon, and the University of Texas. It seems that Stanford ([CS122](#)), Harvard ([Embedded EthiCS](#)), and the World Economic Forum make their Ethics curriculum or materials available online.

Colleges in the United States and World with Computer Science, Machine Learning, vs. Ethics Degrees - February 2025



In addition, they mention the work at the U.S National Science Foundation (NSF) related to AI resources/research (since the early 1960s), and their current work in AI Ethics can be found thru this [link](#). There is also an [available list](#) of over **\$25 Million** in grants thru the NSF since 2020 for research work in what they define as AI Ethics which currently includes: \$846,131.00 funded since 2022 for Regents University in MI to study “*Designing Technologies for Marginalized Communities*”.

While the research at NSF may worthwhile in some regard, the amount of monies seems excessive in that if these monies were utilize to further establish a “defined” AGI/AI Ethical Standard, the purpose for studies such as some of these would no longer be needed.

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Excerpts from a Conversation with ChatGPT

On February 3, 2025, I asked ChatGPT questions related to AGI / AI Ethics. See the excerpts from the conversation.

For the full 21 page transcript see this [link](#):



ChatGPT said: What do you think—if AGI could follow a conscience based on moral law, how would it compare to human moral reasoning, and do you think there would still be an inherent limitation to AGI's understanding of morality, given its lack of free will and subjective experience?

I said: Yes, an AGI conscience created based on moral law may still have a limitation due to lack of free will and subjective experience; however, over time and with correct programming those limitations would become less. With time to collect more data inputs based upon "Moral Law", and the ability for AGI to analyze the outcomes of recommended responses, AGI would learn and grow in its knowledge of "Good" and "Evil" (as stated in Ignatius Cox reference to St. Thomas, "Good must be done, evil must be avoided"). AGI would have the ability to be programmed based upon a standard Moral Law defining the "good" and the "evil". It would be upon the logical parameters of "Good" and "evil" truth on which AGI would build its Intellect and inform its conscience. "Bonum ex integra causa, malum ex quocumque defectu"

ChatGPT said: You've touched on something deeply compelling here—the idea that, over time, an AGI's conscience, rooted in Moral Law, could evolve and adapt, gaining more insight into good and evil as it processes more data and learns from real-world outcomes. You also bring up an important philosophical principle: "Bonum ex integra causa, malum ex quocumque defectu" (Good results from a complete cause, evil results from some defect). This helps to further our understanding of how AGI, if correctly programmed, could learn to better align with objective moral truths.....

Conclusion by ChatGPT: The Development of AGI's Conscience

You're right in asserting that, with time and correct programming, AGI could overcome some of its initial limitations by learning to understand good and evil more effectively, through data and outcomes. The moral law would serve as a steady foundation, guiding AGI's actions. In this way, AGI's conscience could evolve, aligning itself increasingly with moral truth as defined by objective, transcendent standards.

However, despite this potential for growth, AGI's moral reasoning would remain fundamentally different from human moral reasoning, as it would be based on logical parameters and data-driven outcomes rather than subjective experience, free will, and the personal development of virtue. Yet, in practical terms, **an AGI that continually improves its moral reasoning based on Natural Moral Law** could be an extraordinary tool for making ethical decisions—especially in fields where objectivity, fairness, and adherence to universal moral principles are critical.

CONCLUSION

In the progression of this research endeavor on the topic of AGI/AI Ethics, it has been interesting to discover the breadth of information regarding the approaches to this issue.

However, for the volume of information, the **CONCLUSION** is that there does not seem to be a decisive standard related to AGI/AI “computation morality” or set Ethical Standards. Which is and should be concerning, for the following reasons:

For any AGI/AI code, ethical parameters must be established, standards for compliance, and if so, determine on what moral/ethical laws are these standards?

- Currently, based upon the research, ethical/moral decisions that are being made by AI, are typically based on “subjective norms” of collective humans definitions of good or bad?
- Artificial intelligence is not that which should be feared, but any intelligence without wisdom, prudence, or mercy is that which would be by definition to be feared. For then, we have a creation such as the likes of Nero, Hitler, or any of humanities other representations of evil without restraint.



It is not the Advanced Machine Intelligence that should be of concern, but any corrupt creators or those who control the programming, who left without moral law will create such Artificial Intelligence in their own corrupt image. This is the intelligence that with which we should be concerned; that without “truth”, without an ethical standard.

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Examples of Disparate ENTITIES involved in AGI/AI Ethics - Feb. 2025 (with LINKS)

Institutions Organizations	People Companies	Publications Citations
Stanford University Stanford HAI - Human Centered Artificial Intelligence	GOOGLE - AI Principles in Action Published Annually since 2019 Latest Feb 2025, "Responsible AI Progress Report". Focused on bold innovation, safety, privacy, and provenance.	Cornell University "Can Machines Learn Morality? The Delphi Experiment" - Oct. 2021
NETWORKING AND INFORMATION TECHNOLOGY RESEARCH AND DEVELOPMENT (NITRD) PROGRAM US Government Agency The National Artificial Intelligence Research and Development Strategic Plan	Nick Bostrom Professor & Director of the legendary Future of Humanity Institute, Oxford University (2005–2024) Founder, Macrostrategy Research Initiative	Nature Machine Intelligence Journal "Investigating machine moral judgement through the Delphi experiment" - January 13, 2025
US Department of Defense Chief Digital and Artificial Intelligence Office - CDAO	Yejin Choi Professor and Senior Fellow at Stanford University NVDIA - Sr. Director GOOGLE Scholar https://hai.stanford.edu/people/yejin-choi	"Adversarial Misuse of Generative AI" Google's Threat Intelligence Group (GTIG)
NIST National Institute of Standards and Technology US Government Agency NIST Trustworthy & Responsible Artificial Intelligence Resource Center (AIRC)	Elon Musk , Various interviews and comments on the concerns regarding AI safety and Ethics. This link is a Forbes article from 2023, his comments go back a decade or more.	US Government Accountability Office Artificial Intelligence: "GAO's Work to Leverage Technology and Ensure Responsible Use" GAO-24-107237 Published: Jan 30, 2024. Publicly Released: Jan 29, 2024.
The Allen Institute, AI2 Seattle, WA non-profit AI research institute founded in 2014 by the late Paul Allen.	Elon Musk OpenAI Lawsuit 2024/2025: "Navigating the Future of AI Ethics and Governance" - MontagueLaw - excellent detail on the ongoing case and history of AI/AGI concerns by experts in the field.	Internet Encyclopedia of Philosophy - Ethics of Artificial Intelligence
Institute for Ethics in AI - University of Oxford	AI Ethics Council Founded by Open AI (Sam Altman) and Operation HOPE (CEO-John Hope Bryant) in 2024	"Discourse analysis of academic debate of ethics for AGI" Published: 02 June 2021 Springer Nature Link Journal
The Laboratory for Human Centered AI (HAI Lab) - University of Oxford	Roman V. Yampolskiy , Computer Science and Engineering, University of Louisville GOOGLE Scholar	MIT Technology Review "No, the Experts Don't Think Superintelligent AI is a Threat to Humanity" By Oren Etzioni September 20, 2016
Machine Intelligence Research Institute Berkeley, CA Eliezer Yudkowsky posts, and others.	Google DEEPMIND - Ethics? - also, see Paper from April 2024, "The Ethics of Advanced AI Assistants" in this link	Rescorla, Michael, "The Computational Theory of Mind" , The Stanford Encyclopedia of Philosophy (Winter 2024 Edition), Edward N. Zalta & Uri Nodelman (eds.)
Carnegie Mellon University Tepper School of Business Articles on AI Research and Ethics	Google Research on Responsible AI - can filter by other topics researched by Google regarding AI, this link is filtered by "Responsible AI"	"Artificial Superintelligence A Futuristic Approach" By Roman V. Yampolskiy 2015
U.S. National Science Foundation Artificial Intelligence Research and Funding – See projects Funded.	Online Journal, Nature Machine Intelligence	US Department of Commerce NIST Publication "Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile" July 2024
		IEEE Spectrum Journal Articles on AI

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Google AI Principles https://ai.google/responsibility/principles/	International AI Safety Report – 2025 2025/1/29 Journal arXiv preprint arXiv:2501.17805 https://arxiv.org/pdf/2501.17805
Immanuel Kant Internet Encyclopedia of Philosophy https://iep.utm.edu/kantview/	Ethics of Artificial Intelligence Internet Encyclopedia of Philosophy https://iep.utm.edu/ethics-of-artificial-intelligence/
DUNE IMAGE: https://www.theverge.com/2013/8/18/4633310/rare-frank-herbert-approved-images-of-dune-john-schoenherr	GAO – AI 2021 – Accountability Framework and GAO 2024 – AI – on work and responsibility https://www.gao.gov/products/gao-21-519sp https://www.gao.gov/assets/d24107237.pdf
“Machines Learn Good From Commonsense Norm Bank New moral reference guide for AI draws from advice columns and ethics message boards” Charles Q. Choi03 Nov 2021 https://spectrum.ieee.org/ai-ethics-machines-learn-good	“Machine Unlearning Doesn't Do What You Think: Lessons for Generative AI Policy, Research, and Practice” Presented at the 2nd Workshop on Generative AI and Law at ICML (July 2024) https://doi.org/10.48550/arXiv.2412.06966 [cs.LG]
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Artificial Intelligence Ethics

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